

Product Requirements Document (PRD)

Ledger & Credit-Debit Management System

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Executive Summary

Small and medium enterprises (SMEs) frequently struggle with manual bookkeeping, tracking credit extensions, reconciling payments, and notifying customers of outstanding balances. The Ledger & Credit-Debit Management System provides a premium, offline-first desktop application that replaces traditional paper records with a secure database, automates mathematical running balance calculations, generates visual trends, and dispatches statements via native SMTP email and native WhatsApp integration.

By relying on a local SQLite database, the system operates completely offline, safeguarding business ledger directories from remote outages while enforcing multi-user authentication, secure local data backups, and PDF security policies.

1. Product Goals & Target Users

1.1 Core Business Goals

- **Speed & Efficiency:** Enable instant transaction logging with keyboard-driven autocomplete customer fields.
- **Data Security:** Protect transaction logs via secure authentication, password-protected statements, and local database zip archives.
- **Business Intelligence:** Provide instant visibility into cash-flow and balances through a dashboard loaded with financial KPI metric cards and trending charts.
- **Automated Workflows:** Automatically construct and send daily closing summary reports via email and WhatsApp upon application shutdown.

1.2 Target User Profiles

Shopkeepers & Retailers: Non-technical users who require an intuitive desktop app to record daily credits and debits without navigating complex menus. Double-clicking customer rows must instantly display chronicled customer ledgers.

Independent Distributors: Power users who manage dozens of client accounts and require rapid search, live balance filtering, and quick exports of PDF and Excel statements for monthly reconciliations.

Accountants & Bookkeepers: Users who verify transaction histories, calculate running customer balances, and restore database snapshots during audits or system transitions.

2. Functional Requirements

2.1 Dashboard & Analytics Screen

The Dashboard serves as the primary navigation landing screen, offering high-level business intelligence aggregates:

- **KPI Metric Cards:** Display four live aggregates: Total Outstanding Balance (sum of all customer balances), Today's Credit (sum of today's payments), Today's Debit (sum of today's credit sales), and Customer Count (total active records).
- **Matplotlib Trend Chart:** Visually charts daily/weekly credit and debit ratios dynamically, embedded directly within the CustomTkinter frame.
- **Recent Transactions:** A scrollable, clean table highlighting the 10 most recent transactions with client names and colorized transaction types.

2.2 Customer Management Directory

- **CRUD Operations:** Support full creation, reading, updating, and soft/hard deletion of customer files containing names, phone numbers, addresses, opening balances, and notes.
- **Live Filtering:** Real-time, case-insensitive search by name or phone number that dynamically filters the directory grid.
- **Drill-Down:** Double-clicking a customer row redirects the user to that client's Ledger View.
- **Cascading Deletion:** Deleting a customer must automatically trigger cascading SQL deletes across transaction registries and security settings, maintaining database cleanliness.

2.3 Transaction Management Entry

- **Transaction Types:** Support Credit (client payment), Debit (client purchase on credit), and Daily Charge (recurring or flat service fee) options.
- **Customer Autocomplete:** A text-entry field that searches the customer registry and autocompletes matching names as the user types.
- **Data Validations:** Form validations enforcing non-negative numeric amounts, required customer references, and transaction dates.
- **Payment Modes:** Associate transactions with configured modes (Cash, UPI / Online, Card, Bank Transfer, or Cheque) for auditing.

2.4 Reports Engine & Document Exports

- **Ledger Statement Export:** Compile a customer's chronicled transaction history into a premium styled ReportLab PDF or auto-fitted OpenPyXL Excel spreadsheet. PDF files must employ the NumberedCanvas protocol to render page numbers dynamically.
- **Date-Range Reports:** Filter transaction databases by custom start and end dates, customer IDs, or transaction types, presenting results in a tabular format.

2.5 Security, Backup & Restore

- **Password Security Policies:** Enforce complexity validation on the owner password (minimum 8 characters, capital/lowercase letters, numbers, and special symbols).
- **XOR Obfuscation:** Local configuration settings (settings.json) must store sensitive keys (e.g. SMTP passwords) obfuscated via repeating XOR algorithms with salt-keys, preventing plaintext scanning.
- **PDF Encryption:** If PDF Security is enabled in Settings, the export service must post-process generated statements with AES-128 user/owner encryption, prompting the user for a password upon opening.
- **Database Safety Clones:** Let users trigger manual backups. Perform automated backup sweeps before executing a database restoration archive (Pre-Restore), guaranteeing a recovery path.

2.6 Asynchronous Background Notifications

- **Thread-Isolated Boot Alerts:** System tray startup and close notifications run in separate daemon threads, ensuring instant GUI boot times.
- **Daily Closing Summary PDF:** Generated on application close and saved automatically at reports/YYYY-MM-DD/daily_summary.pdf.
- **Secure SMTP Email:** Transmits the daily summary text and generated PDF directly to the configured recipient email address on exit.
- **WhatsApp Desktop Integration:** Triggers on application shutdown. Uses Win32 user32 APIs to focus the native WhatsApp Desktop App, passes the recipient's phone number, pre-fills the daily summary text, copies the summary PDF to the clipboard, and simulates keystrokes to dispatch the message.

3. Non-Functional & Operational Requirements

Performance & Responsiveness: The app's startup sequence, database migrations, and tray updates must complete within 2 seconds. Heavy operations, including PDF building, SMTP transmission, and tray balloon triggers, must occur on thread-isolated pipelines.

Security & Privacy: All business directories, records, configurations, and ledger databases are stored strictly locally on the user's host computer. Passwords must be hashed via PBKDF2 with a random 16-byte salt (100,000 iterations) for security verification.

UI Design Aesthetics: Built using CustomTkinter with modern dark and light theme styles. Uses steel-blue headers, zebra-striped tables, dynamic metrics cards with HSL-tailored colors, and professional typography.